FTA-J1939

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Distributed Fleet Telemetry Aggregator

Internet of Things / Commercial Fleet Edge | Rev A2 - Architecture, Atomic Requirements, Interfaces & Verification Baseline

Mission profile. The aggregator acquires, filters, compresses and buffers heavy-vehicle CAN/J1939 telemetry for commercial trucking and logistics fleets. The released architecture is intentionally read-only at the vehicle-network boundary so telemetry services cannot command vehicle dynamics. Cellular-cost reduction is achieved through rules-based edge analytics and store-and-forward behavior, while missing coverage or backend availability is treated as a communications limitation rather than a reason to disturb vehicle networks.

Engineering baseline. The product uses a Cortex-A compute platform, multiple isolated CAN/J1939 receive interfaces, persistent eMMC storage with wear-leveling, fleet-specific filtering/compression rules and TLS 1.3 mutual authentication to a cloud backend. The supplied 9-36 V commercial-vehicle power statement is retained as the normal input baseline; detailed crank, surge and load-dump waveforms remain program-controlled parameters. High-shock cab mounting, automated fleet-rule flashing and 15-year lifecycle intent are closed through production and field evidence rather than assumed generic limits.

Definition	Controlled product interpretation
Product class	Read-only heavy-vehicle telemetry edge gateway for commercial trucking and heavy-duty logistics fleets.
Safety / assurance	Read-only CAN/J1939 interfaces are intended to prevent interference with vehicle dynamics; runtime data loss is reported as telemetry quality, while release evidence gaps block the affected configuration.
Lifecycle mission	15-year lifecycle intent with eMMC wear monitoring, controlled software/rule updates, credential maintenance, field diagnostics and supplier-change regression.
Primary architecture	Cortex-A compute, isolated multi-channel CAN/J1939 acquisition, local eMMC store-and-forward buffering, rules-based filtering/compression, cellular backhaul and TLS 1.3 mutual authentication.
Release philosophy	Vehicle bus authority, telemetry semantics, storage endurance, cybersecurity, environment and manufacturing evidence are configuration controlled; missing evidence is a release-gate issue, not a fictitious runtime state.

Future vehicle platforms, transceivers, storage devices, modem variants, fleet-rule packages and cloud services are supported only through explicit interface parameters, controlled configuration identities and targeted regression when assumptions change.

PRODUCT GOVERNANCE, REQUIREMENT QUALITY & DATA OWNERSHIP

Product requirements separate product-owned telemetry behavior from fleet policy, cloud-service behavior, vehicle-network ownership and installation assumptions. The released configuration identifies which CAN/J1939 channels are observed, which signal dictionaries and fleet rules are active, how timestamps are formed, how data is filtered or compressed, and where the product-owned retention boundary ends. A throughput or storage claim is valid only for a named traffic profile, rule set, compression mode, cellular condition and power state.

Common release evidence includes hardware revision, software image, fleet-rule package, signal dictionary, certificate state, storage configuration, modem profile, test instrumentation and environmental condition. Parameters not fixed by the program remain TBC with an owner and closure method. Detected runtime faults can trigger buffering, data-quality flags or service restart; missing qualification, cybersecurity, manufacturing or supplier evidence instead keeps the affected configuration outside production equivalence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-GOV-001	Each product requirement shall trace to a mission need, interface obligation, risk control or lifecycle objective and to objective verification evidence.	Bidirectional trace and DVP coverage audit.
FTA-GOV-002	Telemetry performance claims shall identify the traffic profile, rule package, timestamp boundary, storage mode and backhaul condition used for verification.	Controlled performance matrix and test-record review.
FTA-GOV-003	Unclosed qualification, manufacturing, cybersecurity or supplier evidence shall block the affected configuration from production release.	Release-gate and configuration-status audit.

Verification, margin and future-change rule

A change to data ownership, fleet-rule semantics, vehicle-interface hardware, storage layout, modem profile or cloud contract reopens only the analyses and tests whose assumptions are invalidated. Generic fleet planning values shall not be promoted into released product limits without owner-approved evidence.

DEPLOYMENT BOUNDARY, READ-ONLY AUTHORITY & EXPLICIT EXCLUSIONS

The aggregator is an observation and telemetry product, not a vehicle-control ECU. Its vehicle-facing channels acquire CAN/J1939 traffic without intentional transmission of commands, address claims or dominant bits onto the observed buses. The system boundary includes isolated vehicle interfaces, compute, local storage, cellular or Ethernet backhaul, secure credential storage and service diagnostics. Fleet backend services may request configuration or software changes, but those requests cannot create a vehicle-network transmit path that is absent from the released hardware architecture.

Installation-owned items include vehicle connector pinout, harness length, grounding, fuse allocation, antenna placement, external cellular coverage and cloud availability. The product shall distinguish loss of observability from a vehicle fault: a disconnected bus, failed isolator or saturated capture path may reduce telemetry completeness, but it shall not be represented as a vehicle-dynamics condition unless the input data explicitly supports that interpretation. Any diagnostic access to the gateway remains outside vehicle-control authority.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-BND-001	The released hardware shall prevent software or cloud services from intentionally transmitting CAN/J1939 frames onto read-only vehicle-network channels.	Schematic review plus negative transmit-path and pin-fault test.
FTA-BND-002	Each external interface shall define ownership, direction, data validity, timeout behavior and installation dependencies.	ICD and deployment-boundary review.
FTA-BND-003	Loss of telemetry acquisition shall be reported as an aggregator data-quality condition unless vehicle fault semantics are explicitly defined by the signal contract.	Fault-injection and diagnostic-mapping test.

Verification, margin and future-change rule

Connector, harness, isolator, transceiver or service-interface changes require renewed authority and disturbance analysis. An installation workaround cannot be credited to preserve read-only behavior unless it is configuration controlled and verified at the deployed interface.

TELEMETRY MISSION, SOURCE MAPPING & TIMESTAMP SEMANTICS

The product mission is to aggregate, filter and compress heavy-vehicle CAN/J1939 data for fleet use while preserving enough provenance to interpret the resulting telemetry. Every configured signal or frame class is associated with a source channel, decoding rule, timestamp convention, scaling metadata and retention priority. Raw-frame capture, decoded-signal telemetry and event snapshots are treated as distinct data products because they have different completeness, storage and reconstruction expectations.

Timestamp semantics are explicit at the acquisition boundary. The system records whether time represents frame reception, decoded-signal update, rule evaluation or cloud transmission. When absolute time is unavailable, local monotonic ordering may be retained without pretending that UTC is valid. Fleet analytics can therefore distinguish a stale signal, a delayed uplink and an unknown absolute timestamp instead of collapsing those conditions into one generic communication fault.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-DAT-001	Every released telemetry item shall identify source channel, source frame or signal, decoding version, timestamp meaning and retention class.	Signal-dictionary and schema audit.
FTA-DAT-002	Loss of absolute time shall not invalidate local event ordering when a qualified monotonic clock remains available.	Time-loss and recovery test with buffered events.
FTA-DAT-003	Raw, decoded and derived telemetry products shall carry sufficient metadata to prevent one product type from being mistaken for another.	Backend ingestion and schema-validation test.

Verification, margin and future-change rule

DBC/signal-dictionary, J1939 SPN mapping, timestamp source or cloud schema changes reopen mapping and correlation evidence. Previously buffered data shall remain associated with the decoding and rule versions that produced it.

SYSTEM ARCHITECTURE & VEHICLE-TO-CLOUD DATA FLOW

The architecture decomposes the gateway into protected power entry, isolated CAN/J1939 acquisition, Cortex-A compute, volatile buffering, eMMC persistent storage, fleet-rule execution, compression/encoding, cellular backhaul, secure identity services and diagnostics. Data crosses each boundary through a named queue or API with defined ownership, capacity and error behavior. Capture and persistence are separated from uplink so intermittent cellular service does not immediately destroy locally acquired evidence.

Startup sequencing is ordered around power integrity, storage mount, configuration validation, credential readiness, vehicle-interface health and time-service availability. A partial restart of the modem or cloud client does not reset vehicle acquisition unless the shared resource is actually required. Conversely, corruption of the active rule package or storage metadata is detected before the product silently resumes with unknown filtering behavior. Reset reasons and recovery epochs are retained for field diagnostics.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-ARC-001	The released block diagram shall identify power, isolation, acquisition, compute, storage, security and backhaul trust boundaries.	Architecture and interface review.
FTA-ARC-002	Vehicle acquisition shall remain decoupled from temporary backhaul loss while configured local storage capacity remains available.	Cellular-outage store-and-forward test.
FTA-ARC-003	A service restart shall not replay pre-restart telemetry as newly acquired data without preserving its original capture timestamp and epoch.	Service-restart and queue-epoch test.

Verification, margin and future-change rule

Compute-module, storage, modem, queue topology or process-ownership changes require targeted startup, buffering and failure-containment regression. Architectural equivalence shall be demonstrated at the affected boundary rather than assumed from nominal throughput.

9-36 V COMMERCIAL-VEHICLE POWER ENTRY, TRANSIENTS & STARTUP

The released product input envelope is 9-36 V for the commercial-vehicle installation. Crank, reverse-polarity, surge and load-dump pulse shapes are controlled by the approved vehicle interface specification because their amplitude, duration and source impedance are installation dependent. The power entry therefore includes fuse coordination, reverse protection, surge clamping, input filtering, bulk energy storage and supervised conversion to logic, isolation and modem rails.

Reset behavior is part of data integrity. Brownout shall be detected early enough to stop new persistent writes or transition the storage stack into a controlled power-fail path. The design records whether a shutdown was commanded, undervoltage-driven or abrupt so post-restart diagnostics can distinguish expected power cycling from field wiring or transient problems. Hold-up energy is justified against the actual storage and modem shutdown sequence rather than by an unverified capacitor value.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-PWR-001	The released 9-36 V input variant shall define its normal range and separately control each qualified transient and reverse-polarity condition.	Power-interface specification and transient DVP.
FTA-PWR-002	Brownout detection shall protect configuration and persistent-storage integrity before rail collapse violates device write requirements.	Programmable ramp-down and repeated power-interruption test.
FTA-PWR-003	A restart after abnormal power loss shall validate storage, active rules and credential state before normal telemetry service resumes.	Power-cut recovery and filesystem/configuration audit.

Verification, margin and future-change rule

Changes to TVS devices, DC/DC stages, bulk capacitance, eMMC power-fail requirements or vehicle feeder assumptions reopen transient, hold-up, EMC and restart verification. A generic “heavy vehicle” pulse shall not substitute for the released installation profile.

CORTEX-A COMPUTE PLATFORM, MEMORY & RESOURCE RESERVE

The Cortex-A platform executes acquisition services, rule evaluation, compression, persistent buffering, security clients and backhaul communication. CPU, memory and I/O capacity are budgeted so a cellular retry storm, storage maintenance operation or diagnostic session cannot starve the vehicle-capture path. Memory ownership is explicit for kernel buffers, CAN receive queues, event snapshots, compression workspaces and upload staging areas.

Health monitoring captures reset cause, watchdog events, memory pressure, CPU saturation and repeated service failures. The platform treats sustained resource exhaustion as a data-quality problem: lower-priority analytics or upload work can be shed before frame acquisition is abandoned. Production-equivalent benchmarking uses the released kernel, compiler, storage device, rule package and traffic profile because a synthetic CPU score does not establish end-to-end telemetry capacity.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-CPU-001	The released compute configuration shall maintain defined CPU and memory reserve for vehicle acquisition under the qualified worst-case concurrent workload.	Stress test with capture, compression, storage and cellular activity.
FTA-CPU-002	Resource exhaustion shall be detected and shall preferentially shed lower-priority work before corrupting high-priority acquisition queues.	CPU/memory pressure fault test.
FTA-CPU-003	Production-equivalent performance evidence shall use the released BSP, memory, storage and fleet-rule configuration.	Configuration-linked benchmark audit.

Verification, margin and future-change rule

Processor stepping, DRAM density, kernel, compiler, scheduler or storage-driver changes reopen the resource and latency measurements they can affect. Supplier claims alone are insufficient where memory pressure or interrupt latency determines capture completeness.

ISOLATED MULTI-CHANNEL CAN / J1939 FRONT ENDS

Each vehicle channel includes a defined connector path, protection network, transceiver, galvanic or functional isolation as allocated, and receive interface to the compute platform. The design controls input capacitance, leakage, common-mode range and unpowered behavior because a telemetry device can disturb a healthy bus even when its software never transmits. Unused or unsupported channel pins are left in a deterministic non-driving condition.

Channel identity is preserved from connector to data record so fleets can distinguish chassis, powertrain, body or auxiliary networks without relying on order-of-enumeration. Diagnostics detect loss of transceiver power, isolation supply failure, stuck receiver output and persistent bus inactivity. A bus that is simply quiet is not automatically declared faulty; inactivity interpretation remains channel- and fleet-specific.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-CAN-001	Each read-only CAN/J1939 channel shall meet the released bus-loading, leakage and unpowered-disturbance limits for its installation.	Electrical characterization at min/max supply and temperature.
FTA-CAN-002	Channel identity shall remain deterministic across restart, software update and hardware variant changes.	Connector-to-record trace test.
FTA-CAN-003	A failed acquisition channel shall be isolated from other channels and shall not prevent unaffected interfaces from continuing telemetry capture.	Single-channel power/transceiver fault injection.

Verification, margin and future-change rule

Transceiver, isolator, common-mode choke, protection network, connector or PCB-stack changes require renewed loading, EMC and fault-containment evidence on the affected channels. Logical read-only intent does not waive physical-layer disturbance verification.

READ-ONLY COUPLING & BUS-DISTURBANCE CONTAINMENT

Read-only behavior is enforced by architecture rather than policy alone. The vehicle-facing transmit capability is removed, permanently disabled or hardware-gated so an application defect, compromised service or unexpected pin state cannot intentionally generate vehicle commands. The design review traces every potential path from processor GPIO, transceiver TXD, boot strap or test point to the CAN bus and documents why it cannot create a dominant-bit sequence in the released configuration.

Electrical failure cases are also considered. Shorts across isolation, damaged transceivers, unpowered bias networks and connector faults are analyzed for their ability to load or clamp the bus. Where a single hardware fault can still disturb the network, the resulting dependency is documented in the risk analysis and validated against the installation protection concept. Manufacturing tests confirm the receive-only population and configuration before shipment.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-RO-001	No software-controlled path shall be capable of intentionally driving a released read-only vehicle-network channel.	Schematic/netlist inspection and adversarial software test.
FTA-RO-002	Single-component faults that can electrically disturb the bus shall be identified and evaluated against the installation risk controls.	FMEDA/FMEA-style interface analysis plus fault insertion where practical.
FTA-RO-003	End-of-line testing shall detect an incorrectly populated or enabled transmit-capable path on vehicle-facing channels.	EOL read-only enforcement test.

Verification, margin and future-change rule

Any change to transceiver mode control, boot pin behavior, production test access or processor pin mux reopens read-only enforcement analysis. A cybersecurity claim cannot replace the physical proof that the vehicle interface lacks unauthorized transmit authority.

CAN / J1939 MESSAGE INGEST & TRANSPORT-PROTOCOL HANDLING

The acquisition stack accepts standard CAN/J1939 traffic according to the configured vehicle-network profile and preserves frame identifiers, channel, DLC, payload, reception status and timestamp. J1939 decoding is separated from raw capture so unknown PGNs or proprietary payloads can still be retained when policy allows. Multi-packet messages are reconstructed only when the required fragments, ordering and timing conditions are satisfied.

The product does not participate in address claiming on read-only channels. Transport-protocol observations such as BAM or connection-managed exchanges are interpreted passively; missing packets, overlapping sessions or inconsistent source addresses are flagged rather than repaired with invented data. The fleet rule engine consumes decoded values only after parser validity checks, preventing malformed or partial transport messages from masquerading as complete application data.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-J19-001	Raw capture shall preserve enough CAN/J1939 metadata to reconstruct the observed frame sequence within the released buffering limits.	Recorded-trace replay and byte-level comparison.
FTA-J19-002	Multi-packet J1939 payloads shall be published as complete only after the configured reconstruction validity checks pass.	BAM/transport-protocol loss, reorder and overlap tests.
FTA-J19-003	The aggregator shall not originate J1939 address claims or transport-protocol control frames on read-only channels.	Network analyzer verification during startup and runtime.

Verification, margin and future-change rule

Parser, DBC/SPN database or J1939 transport implementation changes reopen only the affected decode and reconstruction test suites. Raw-capture fidelity provides the reference needed to distinguish parser changes from vehicle-data changes.

TIMESTAMPING, CLOCK ARCHITECTURE & EVENT ORDERING

Telemetry value is reduced if capture time cannot be interpreted. The architecture therefore maintains a monotonic local clock for ordering and, when available, a qualified absolute-time source for correlation across vehicles and cloud systems. The record format identifies clock quality so data captured during uncertain absolute time is not silently labeled with precise UTC. Clock step, slew and restart behavior are handled without reversing local event order.

Timestamp insertion occurs as close to the receive boundary as practical and before variable processing such as rule evaluation, compression or cellular queuing. For derived signals, the system distinguishes source-frame time from derivation time. Cross-channel correlation tests exercise simultaneous traffic and high CPU load to quantify skew introduced by interrupt servicing and software buffering.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-TIM-001	Every persisted telemetry record shall identify the capture timestamp and time-quality state associated with that timestamp.	Record-format inspection and time-source fault test.
FTA-TIM-002	Loss or correction of absolute time shall not reverse monotonic local event ordering.	Clock-step, holdover and restart correlation test.
FTA-TIM-003	Release evidence shall characterize timestamp skew across active vehicle channels under the qualified peak traffic load.	Synchronized stimulus and trace correlation.

Verification, margin and future-change rule

Clock source, kernel timestamping method, driver, interrupt policy or virtualization changes reopen time-correlation evidence. Cloud receipt time shall not be substituted for vehicle capture time in requirements that depend on event sequence.

EDGE FILTERING RULE ENGINE & POLICY VERSIONING

Rules-based filtering reduces cellular cost by deciding which frames or decoded signals are retained, sampled, aggregated, event-triggered or discarded. Rule execution is deterministic for a given input sequence, rule package and configuration. Safety-critical vehicle behavior is not controlled by these rules because the aggregator is read-only; however, poor rules can still remove evidence needed for fleet diagnostics, so rule ownership and change approval are configuration-controlled.

Each rule package carries version, fleet/vehicle applicability, dependency on signal-dictionary version and rollback identity. Invalid syntax, missing dependencies or unsupported operators prevent activation of the new package while the previous qualified package remains available. Rule evaluation records enough diagnostics to determine whether data was absent at acquisition, rejected by validation or deliberately filtered by policy.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-RUL-001	The active fleet-rule package shall be cryptographically or configuration-integrity checked before activation and shall have an explicit applicability identity.	Provisioning and corrupted-package test.
FTA-RUL-002	Filtering decisions shall distinguish missing input data from policy-driven suppression.	Rule-engine diagnostic and trace review.
FTA-RUL-003	A failed rule update shall preserve or restore the last qualified rule package without enabling a partially applied configuration.	Interrupted update and rollback test.

Verification, margin and future-change rule

Operator semantics, rule compiler/interpreter, signal dictionary or fleet applicability changes require targeted golden-trace regression. Optimization of rule execution may not change the released filtering result without a controlled behavior revision.

COMPRESSION, ENCODING & TELEMETRY SCHEMA

Compression is used to reduce backhaul cost without obscuring data provenance. The product distinguishes lossless encoding, controlled down-sampling, delta encoding and statistical aggregation because each changes reconstructability differently. The telemetry schema records units, scaling, signedness, missing-value representation and version so cloud services do not infer meaning from field position alone.

Compression is evaluated against worst-case signal dynamics and event bursts, not only steady cruising data. When a lossy reduction is permitted by fleet policy, the acceptable information loss is defined by the owning use case rather than a generic compression ratio. Decoder compatibility is verified across supported backend versions, and malformed compressed blocks are rejected without corrupting adjacent records.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-CMP-001	Every compression mode shall declare whether the resulting telemetry is lossless, sampled or aggregated and shall identify the corresponding reconstruction semantics.	Schema and encoder/decoder contract review.
FTA-CMP-002	Compressed blocks shall include integrity and version information sufficient to detect incompatible or corrupted payloads before ingestion.	Corruption and version-mismatch test.
FTA-CMP-003	Any approved lossy reduction shall have use-case-owned acceptance criteria for the information removed.	Use-case review plus representative trace comparison.

Verification, margin and future-change rule

Codec, schema, scaling or down-sampling changes reopen decoder interoperability and use-case fidelity tests. A higher compression ratio is not automatically an improvement if it removes transient evidence required for diagnostics.

OFFLINE BUFFERING & STORE-AND-FORWARD BEHAVIOR

The gateway is expected to operate through intermittent cellular coverage. Captured data is therefore written to bounded local queues with explicit priority, retention and eviction policy. Critical event snapshots can be protected from ordinary rolling telemetry where fleet policy requires it, while lower-value data may be dropped first when persistent storage approaches its configured limit. Queue state is recoverable after restart without presenting old data as newly captured.

Connectivity recovery does not permit uncontrolled upload bursts that starve acquisition or saturate the modem. Store-and-forward scheduling limits CPU, storage read bandwidth and backhaul duty so new vehicle data remains observable. Cloud acknowledgements are used only where the protocol defines them; absence of acknowledgement cannot cause indefinite duplicate retransmission without bounded retry and deduplication metadata.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-BUF-001	Offline buffering shall use a released priority and eviction policy that prevents uncontrolled storage exhaustion.	Long-duration outage and capacity-exhaustion test.
FTA-BUF-002	Queued records shall retain original capture timestamps and unique ordering metadata across restart and later upload.	Power-cycle store-and-forward correlation test.
FTA-BUF-003	Backlog upload after connectivity recovery shall be rate bounded so vehicle acquisition retains its required resource reserve.	Recovery-from-outage stress test.

Verification, margin and future-change rule

Retention policy, queue sizing, record format, upload acknowledgement or storage partition changes reopen outage-duration and recovery-load verification. Field policy changes that reduce protected event retention require explicit fleet-owner approval.

EMMC STORAGE ARCHITECTURE, PARTITIONS & POWER-LOSS INTEGRITY

eMMC provides persistent buffering, software/configuration storage and service logs. These functions are separated logically so an abnormal growth in telemetry backlog cannot consume the space required for boot, rollback or critical configuration. Filesystem and database choices are evaluated for abrupt power interruption, write amplification and recovery time. Where device health information is available, it is captured as lifecycle evidence rather than treated as an exact remaining-life predictor.

Persistent writes are grouped and journaled according to the selected storage architecture, with explicit treatment of partially written records and metadata. On startup, the system validates the active filesystem, rule package and queue index before resuming upload. Corruption of one telemetry segment is contained so unaffected segments remain recoverable. Storage service shall expose saturation, bad-block, mount and recovery diagnostics to fleet maintenance.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-EMM-001	Boot/configuration storage shall be protected from telemetry-backlog growth by released partition or quota controls.	Storage-capacity and quota-exhaustion test.
FTA-EMM-002	Abrupt power interruption shall not cause a partially written telemetry record to be accepted as valid after restart.	Repeated power-cut and record-integrity campaign.
FTA-EMM-003	Storage health and corruption diagnostics shall identify whether data loss is localized, capacity-related or device-related.	Fault injection and service-diagnostic review.

Verification, margin and future-change rule

eMMC vendor, density, filesystem, journaling mode or partition layout changes reopen power-loss, endurance and recovery testing. Nominal capacity equivalence does not demonstrate equivalent write endurance or power-fail behavior.

CELLULAR MODEM, BACKHAUL & NETWORK-AVAILABILITY BOUNDARY

The cellular subsystem provides cloud backhaul but is not assumed to be continuously available. Modem hardware, SIM/eSIM profile, APN, roaming policy, antenna path and network technology are variant-controlled. The telemetry service separates modem registration state, IP connectivity, authenticated cloud session and application acknowledgement so a failure at one layer is not misdiagnosed as another.

Repeated registration or connection failures use bounded retry and backoff to protect power, thermal and data-capture resources. The modem cannot force the gateway to discard local records solely because coverage is absent. Network changes may alter bandwidth and latency; therefore backhaul performance is characterized independently from vehicle acquisition fidelity. Field diagnostics expose signal quality and session counters without claiming that RSSI alone predicts usable throughput.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-CEL-001	The product shall distinguish modem registration, IP connectivity, TLS session and application-service availability in diagnostics.	Network-layer fault matrix test.
FTA-CEL-002	Backhaul retry behavior shall be rate bounded and shall not starve vehicle acquisition or persistent buffering.	Coverage-loss and retry-storm resource test.
FTA-CEL-003	Loss of cellular service shall not delete buffered data except through the released retention/eviction policy.	Extended outage test with storage audit.

Verification, margin and future-change rule

Modem, carrier profile, SIM/eSIM implementation, APN or antenna changes reopen connectivity, thermal, RF and cybersecurity evidence as applicable. Network-provider availability remains an external dependency and is not represented as product-owned uptime.

TLS 1.3 MUTUAL AUTHENTICATION & SECURE CLOUD CHANNEL

The supplied cybersecurity baseline requires TLS 1.3 with mutual authentication to the cloud backend. The product therefore authenticates both the gateway and intended service endpoint before telemetry or management traffic is accepted on the protected channel. Certificate validation, hostname or service identity, protocol version policy and trust-anchor selection are controlled by the released security configuration rather than by ad hoc application settings.

A TLS failure cannot disable local read-only acquisition or storage protection. The gateway records actionable reason classes such as expired credential, trust failure, time invalidity or transport failure without exposing private keys. Management commands received after authentication still pass authorization and configuration checks; encrypted transport alone does not grant authority to modify fleet rules, software or retained data.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-TLS-001	Cloud sessions carrying telemetry or remote management shall use TLS 1.3 mutual authentication with the released trust configuration.	Protocol capture and certificate-authentication test.
FTA-TLS-002	TLS authentication failure shall leave local acquisition and protected buffering available while blocking the unauthenticated cloud session.	Credential-fault and network-isolation test.
FTA-TLS-003	Authenticated transport shall not by itself authorize configuration, software or credential changes.	Role/command authorization negative test.

Verification, margin and future-change rule

TLS library, cipher policy, certificate profile, backend identity or time-validation dependency changes reopen interoperability and failure-handling tests. Security updates that narrow accepted algorithms may be released only with backend compatibility evidence.

SECURE BOOT, SOFTWARE IMAGE CONTROL & UPDATE RECOVERY

Software integrity begins before the telemetry application starts. The boot chain verifies the platform components required by the released security architecture, then validates the operating system and application/configuration package identities before normal cloud management is enabled. Update packages are authenticated and bound to compatible hardware and configuration variants so a valid image for one product option is not silently applied to another.

Update interruption is treated as a recoverable service event rather than a reason to risk unreadable storage or unknown fleet rules. The product maintains a defined rollback or recovery path and preserves diagnostic evidence of the failed attempt. Vehicle acquisition authority remains read-only throughout update states; if acquisition is intentionally paused, the gap is marked in telemetry-quality records so fleet analytics can distinguish maintenance downtime from missing vehicle traffic.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-SEC-001	The released boot chain shall authenticate the software components designated by the product security architecture before normal management services become active.	Secure-boot positive/negative image test.
FTA-SEC-002	Update packages shall be accepted only for compatible hardware/software variants and authorized signing identities.	Variant-mismatch and signature-failure test.
FTA-SEC-003	Interrupted update shall recover to a known software/configuration state and retain an auditable failure reason.	Power-loss-during-update and rollback test.

Verification, margin and future-change rule

Bootloader, signing key hierarchy, partition map or update agent changes reopen boot, rollback and anti-mismatch evidence. Recovery mechanisms shall be tested on production-equivalent storage because failure behavior can depend on device and filesystem details.

KEY / CERTIFICATE LIFECYCLE, SECURE STORAGE & PROVISIONING

Device identity, TLS credentials and any fleet-management authorization material are stored in protected hardware or software facilities appropriate to the released security design. Provisioning associates credentials with unit serial number, hardware identity and manufacturing genealogy. Private keys are not exported through normal service interfaces, and certificate renewal or replacement cannot silently change the unit identity used for backend authorization.

Lifecycle handling covers issuance, activation, renewal, expiry, revocation and decommissioning. The gateway tolerates temporary backend unavailability without weakening credential validation. Time-dependent certificate checks explicitly handle invalid or unknown absolute time. Field replacement of a compute module or secure element follows an identity transfer or reprovisioning process rather than copying credentials as ordinary files.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-KEY-001	Provisioned device credentials shall be bound to the serialized hardware identity and recorded in manufacturing genealogy.	Provisioning database and unit-identity audit.
FTA-KEY-002	Private authentication material shall not be exportable through normal fleet-management or diagnostic interfaces.	Interface penetration/negative-access test.
FTA-KEY-003	Credential renewal, expiry and revocation states shall produce deterministic cloud-access behavior without weakening local telemetry protection.	Lifecycle state-transition test.

Verification, margin and future-change rule

Secure element, key store, certificate authority or provisioning flow changes require renewed identity-binding and recovery evidence. Second-source secure hardware is not production equivalent until key-protection and lifecycle behavior are demonstrated.

FLEET-SPECIFIC RULE CONFIGURATION & REMOTE COMMAND AUTHORITY

Fleet-specific routing and filtering rules are a controlled product input because they determine what data leaves the vehicle and how much is retained locally. The remote-management interface separates read-only status queries, rule updates, software updates and credential operations into distinct command classes with authorization, version and audit requirements. A backend operator cannot gain vehicle-control authority through these services because the vehicle interfaces remain physically read-only.

Configuration changes are staged, validated and activated atomically. Compatibility checks cover signal dictionary, rule-engine version, storage requirements and fleet/vehicle applicability. The product records who or what authorized the change, the resulting configuration identity and rollback status. A malformed or unauthorized command is rejected without disturbing the currently active telemetry configuration.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-CFG-001	Each remote configuration command shall be authenticated, authorized by command class and logged with resulting configuration identity.	Role-based command test and audit-log review.
FTA-CFG-002	Fleet-rule activation shall be atomic with respect to the released compatibility checks.	Interrupted and incompatible configuration test.
FTA-CFG-003	Remote management shall not create or enable transmit authority on read-only vehicle interfaces.	End-to-end authority review and adversarial command test.

Verification, margin and future-change rule

Backend API, role model, fleet-rule packaging or activation mechanism changes reopen authorization and rollback tests. A cloud-side permission change does not remove the need for product-side command validation.

WATCHDOG, HEALTH SUPERVISION & CONTROLLED SERVICE RECOVERY

Health supervision distinguishes failures of acquisition, storage, rule execution, modem, security and management services. A watchdog reset is a last recovery action rather than the only diagnostic. Where possible, isolated services are restarted without discarding healthy capture queues or remounting storage. Repeated restart loops are detected and escalated to a bounded degraded state that preserves diagnostic evidence.

Health status includes CPU/memory pressure, queue depth, storage state, CAN-channel activity, modem session state, credential validity and last successful cloud exchange. These indicators are not converted into vehicle fault claims. The product retains reset cause, software version and active configuration around recovery events so field teams can correlate intermittent telemetry gaps with product behavior.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-HLT-001	The gateway shall supervise acquisition, storage and backhaul services with fault indicators that identify the failed subsystem rather than only a global watchdog event.	Service fault-injection matrix.
FTA-HLT-002	Recoverable service faults shall be contained so unaffected acquisition channels and stored data remain available when architecture permits.	Selective process and peripheral reset test.
FTA-HLT-003	Repeated recovery failures shall be rate bounded and logged with reset cause, software and configuration identity.	Restart-loop and diagnostic-retention test.

Verification, margin and future-change rule

Watchdog source, service partitioning, restart policy or supervisor ownership changes reopen failure-containment evidence. Recovery timing shall be verified with realistic storage and modem states because startup cost is workload dependent.

POWER, THERMAL & CAB-MOUNT ENVIRONMENTAL BASELINE

The gateway is intended for commercial trucking and heavy-duty logistics fleets and shall operate within the cab-mounted electrical, thermal and mechanical environment defined by the vehicle program. The controlled baseline includes the 9-36 V input intent and high-shock chassis packaging; temperature, vibration and ingress limits remain TBC until the installation class is approved.

Thermal design accounts for Cortex-A compute, eMMC write activity, isolated interface losses and cellular transmit peaks. Temperature sensing is placed so sustained thermal stress can be distinguished from transient modem activity. Thermal protection may reduce uplink duty or nonessential analytics before capture integrity is affected, provided the action and resulting data-quality impact are observable. Mounting, airflow and nearby heat sources are treated as installation parameters.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-ENV-001	Each released installation variant shall define the electrical, thermal, shock/vibration and mounting envelope used for qualification.	Installation specification and qualification matrix.
FTA-ENV-002	Thermal protection shall preserve high-priority acquisition where feasible before shedding lower-priority upload or analytics workloads.	Thermal chamber stress test with full workload.
FTA-ENV-003	Temperature-related degradation shall be distinguishable from cellular coverage or storage faults in diagnostics.	Combined thermal/network fault test.

Verification, margin and future-change rule

Enclosure, thermal interface, compute power, modem, mounting location or airflow assumption changes reopen the affected thermal and mechanical evidence. Unspecified environmental values remain controlled parameters rather than inferred automotive defaults.

PART II - VEHICLE BUS LOADING, LEAKAGE & TERMINATION DISTURBANCE

A read-only aggregator still adds electrical loading to every connected vehicle network. The validation therefore measures differential and common-mode input impedance, leakage with the gateway powered and unpowered, connector/ESD network capacitance and any unintended termination contribution. Multi-channel operation is tested simultaneously because shared grounds, isolation supplies or protection structures can create coupling that is absent in a single-channel bench setup.

Disturbance evidence covers normal supply, brownout, startup, shutdown and representative fault conditions. The acceptance limit is installation-specific and remains TBC until the vehicle-network electrical interface is released. The key release principle is that the aggregator shall not consume enough network margin to alter communication behavior or make the vehicle dependent on the telemetry device for termination or bias.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-LDG-001	The aggregator shall not provide intentional vehicle-network termination or bias unless explicitly allocated by the installation design.	Resistance/capacitance inspection and bus measurement.
FTA-LDG-002	Powered and unpowered bus loading shall remain within the released installation limits on every channel.	Impedance/leakage characterization.
FTA-LDG-003	Simultaneous multi-channel operation shall be tested for shared-path coupling that can disturb an otherwise healthy network.	Multi-bus stress and common-mode injection test.

Verification, margin and future-change rule

Protection, choke, isolator, transceiver or PCB-layout changes reopen loading and common-mode measurements. A device that is software read-only can still fail the product requirement if its physical interface degrades bus margin.

ISOLATION BARRIER, COMMON-MODE & ESD / TRANSIENT ROBUSTNESS

Isolation protects the compute and cross-channel architecture from vehicle-network ground differences and transient energy where isolation is allocated. Each barrier is analyzed as a system including transceiver-side supply, digital isolator, creepage/clearance, protection network and return path. Common-mode events are verified for both communication integrity and the possibility of unwanted current through cable shields, chassis or service connectors.

ESD and transient tests capture whether the gateway recovers without corrupting storage or violating read-only behavior. A transient-induced processor reset is not acceptable as the only containment strategy if it can create repeated queue loss or uncontrolled restart. Barrier degradation or isolation-supply failure is diagnosed where practical and contained to the affected channel.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-ISO-001	Each isolated vehicle interface shall meet the released isolation and common-mode requirements for its installation.	Isolation/withstand and common-mode communication test.
FTA-ISO-002	ESD or transient exposure shall not create a sustained vehicle-bus drive condition on read-only channels.	Transient test with bus-state monitoring.
FTA-ISO-003	A failed isolation supply or barrier shall be contained so unrelated channels remain available when physically independent.	Channel isolation fault injection.

Verification, margin and future-change rule

Isolation component, PCB spacing, protection network or chassis return changes require renewed barrier and transient evidence. Qualification data from a component alone does not establish system-level common-mode behavior.

J1939 ADDRESS CLAIM, SOURCE IDENTITY & PASSIVE OBSERVATION SEMANTICS

Heavy-vehicle J1939 networks can change source addresses through address-claim behavior. Because the aggregator is passive, it observes claims and associates decoded data with the source identity visible on the bus without participating in the arbitration. Fleet configuration can map known source identities to logical ECUs, but the product preserves the raw source address and claim context so dynamic changes remain traceable.

Conflicting claims, device replacement or proprietary addressing are not automatically resolved by assumption. The rule engine may apply fleet-specific mappings only when the configured evidence is present. Data from an unknown or changed source can be retained with reduced semantic confidence rather than silently attributed to the previous ECU. This protects downstream maintenance analytics from identity drift after service events.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-ADR-001	The aggregator shall observe J1939 source-address changes without originating address-claim traffic on read-only channels.	Address-claim trace test.
FTA-ADR-002	Decoded telemetry shall preserve raw source-address context when logical ECU identity is derived by fleet mapping.	Source mapping and replacement-ECU test.
FTA-ADR-003	Conflicting or unknown source identities shall not be silently mapped to a previously known ECU.	Conflict and unknown-source validation test.

Verification, margin and future-change rule

Fleet identity tables, parser behavior or source-mapping logic changes reopen address-claim and service-replacement regression. Raw trace retention remains the reference for disputes about logical ECU attribution.

J1939 TRANSPORT PROTOCOL - BAM / SESSION RECONSTRUCTION

Large J1939 payloads may use broadcast or session-managed transport. The passive aggregator reconstructs these messages from observed packets while respecting source, destination, sequence and timeout context. It does not transmit CTS, abort or acknowledgement traffic. Reconstruction state is bounded so malformed or abandoned sessions cannot consume unbounded memory or block newer traffic.

Partial messages can still be useful for raw diagnostics, but they are not exposed as complete decoded application payloads. The telemetry schema distinguishes complete, partial, duplicate and timed-out transport sessions. Concurrent sessions and channel bursts are included in performance testing because transport reassembly can create memory and CPU pressure that ordinary single-frame traffic does not reveal.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-TP-001	Transport-protocol reassembly shall publish a decoded payload as complete only when all required session validity conditions are satisfied.	BAM/session golden-trace replay.
FTA-TP-002	The passive aggregator shall not transmit transport-protocol control or acknowledgement frames on read-only networks.	Bus-analyzer negative transmission test.
FTA-TP-003	Incomplete or maliciously extended sessions shall have bounded memory and timeout behavior.	Session-flood and timeout stress test.

Verification, margin and future-change rule

Transport parser, memory pool, timeout logic or decode schema changes reopen concurrent-session and partial-message tests. Performance evidence shall include representative multi-packet bursts, not only single-frame averages.

DATA LOSS, QUEUE OVERFLOW, CONGESTION & BACKPRESSURE

Every buffering layer has finite capacity: driver receive queues, user-space queues, compression staging, persistent storage and uplink buffers. The product therefore defines overload behavior instead of assuming average traffic. When capacity is exhausted, the loss is measured, classified and exposed with channel and time context. High-priority event data may receive reserved capacity, while lower-priority periodic telemetry can be sampled or dropped according to fleet policy.

Backpressure is not allowed to propagate into the vehicle network because the interface is read-only. Instead, the gateway adapts internal work by reducing analytics, deferring upload or applying configured filtering. Overflow counters are monotonic across service restarts where practical and are included in health telemetry so cloud analytics can separate “vehicle silent” from “gateway dropped data.”

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-CNG-001	Each acquisition and persistence queue shall have a released capacity, priority policy and observable overflow counter.	Peak-load and forced-overflow test.
FTA-CNG-002	Internal congestion shall not create transmit behavior or electrical backpressure on read-only vehicle networks.	Saturation test with bus-state monitoring.
FTA-CNG-003	Dropped-data diagnostics shall identify affected channel, time interval and reason class.	Queue-fault and cloud-report correlation test.

Verification, margin and future-change rule

Queue sizing, scheduling, compression or storage changes reopen overload tests for the affected layers. Increasing buffer depth without reviewing latency, endurance and recovery time is not considered a complete fix.

STALE, DUPLICATE & OUT-OF-ORDER TELEMETRY HANDLING

Vehicle networks, persistent queues and cellular retries can each create data that arrives late or more than once. The aggregator therefore carries capture timestamp, sequence or record identity, source channel and upload epoch so downstream services can detect duplicates and ordering boundaries. Reconnection does not cause old buffered records to be relabeled with current time merely because they are transmitted later.

Signal-derived telemetry uses freshness rules appropriate to the configured source rate. A repeated CAN value is not automatically stale if the source legitimately transmits the same payload; staleness is based on update timing and source semantics. When ordering cannot be proven across independent channels, the product represents uncertainty rather than inventing a total order.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-ORD-001	Buffered telemetry shall retain original capture time and record identity through retry and reconnect cycles.	Offline/reconnect duplicate test.
FTA-ORD-002	Signal freshness shall be determined from configured update semantics rather than value change alone.	Constant-value and missed-update test.
FTA-ORD-003	Cross-channel ordering uncertainty shall be represented explicitly when the time architecture cannot prove a deterministic order.	Multi-channel timing correlation test.

Verification, margin and future-change rule

Record identity, timestamp precision, acknowledgement or deduplication changes reopen reconnect and ordering regression. Backend changes shall preserve the distinction between capture order and cloud-ingestion order.

EVENT TRIGGERING, PRE-TRIGGER & POST-TRIGGER SNAPSHOT BUFFER

Fleet rules may preserve higher-resolution evidence around selected events such as diagnostic codes, threshold crossings or unusual operating combinations. The event subsystem maintains a rolling pre-trigger buffer and configurable post-trigger retention where supported by the released storage budget. Trigger evaluation uses validated input signals and records the rule identity that caused capture escalation.

Simultaneous or repeated triggers are merged, queued or prioritized according to explicit policy so an event storm cannot exhaust all storage. A trigger is not interpreted as a safety decision; it is a telemetry retention decision. Snapshot metadata includes the protected time window, source channels, rule version and any acquisition loss that occurred during the event, allowing downstream engineers to assess evidence completeness.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-EVT-001	Event snapshots shall identify trigger rule/version, capture window, participating channels and data-loss indicators.	Triggered-event trace review.
FTA-EVT-002	Pre-trigger data shall be preserved for the configured duration or shall indicate when insufficient history was available.	Startup and late-trigger test.
FTA-EVT-003	Event storms shall have bounded storage and priority behavior that protects the gateway from uncontrolled retention growth.	Repeated-trigger capacity stress test.

Verification, margin and future-change rule

Trigger semantics, snapshot duration, storage reservation or merge policy changes reopen event completeness and capacity testing. Use-case owners shall approve any reduction of protected pre/post-event evidence.

COMPRESSION FIDELITY & CLOUD RECONSTRUCTION CORRELATION

A compressed telemetry stream is released only when the backend can reconstruct the intended information with known semantics. Golden traces compare vehicle input, local encoded records and cloud-decoded output. This correlation includes units, scaling, signed values, missing markers, timestamp precision and event boundaries. For lossless modes, byte or value equivalence is required after defined metadata transformations.

For sampled or aggregated modes, the test instead demonstrates the approved fidelity metric and clearly identifies unavailable detail. The gateway and cloud decoder versions are configuration linked so a backend update cannot silently reinterpret historical data. Representative traces include ramps, wraparound, bursty counters, diagnostic messages and transport-protocol payloads rather than only smooth analog-like signals.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-RCN-001	Lossless telemetry modes shall reproduce the released decoded values and metadata at the backend for golden input traces.	End-to-end golden-trace comparison.
FTA-RCN-002	Sampled or aggregated modes shall meet use-case-owned fidelity criteria and identify the reduction mode in the schema.	Fidelity analysis and use-case acceptance.
FTA-RCN-003	Gateway encoder and cloud decoder compatibility shall be configuration controlled across software releases.	Version-matrix interoperability test.

Verification, margin and future-change rule

Encoder, decoder, schema or numerical representation changes reopen golden-trace correlation. Historical data compatibility must be assessed explicitly when a backend decoder is replaced.

CLOUD UPLINK QOS, BANDWIDTH BUDGET & CELLULAR-COST CONTROL

The edge analytics objective is to minimize cellular cost without compromising the fleet evidence classes defined by policy. The release therefore budgets average and burst uplink volume by telemetry class, compression mode and duty cycle. Event data, health status, configuration acknowledgements and bulk backlog can use different priorities so a large offline queue does not delay urgent device-health information indefinitely.

Cost control remains separate from availability: the gateway can reduce ordinary telemetry frequency or defer low-priority backlog according to policy, but it does not discard protected records solely because a billing threshold is reached unless that behavior is explicitly owned by the fleet configuration. Upload scheduling accounts for network retries and protocol overhead rather than using compressed payload bytes as the only bandwidth metric.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-QOS-001	The released fleet profile shall define uplink priority and bandwidth policy by telemetry class.	Policy review and traffic-shaping test.
FTA-QOS-002	Backlog upload shall not indefinitely starve higher-priority health or configuration traffic.	Mixed-priority outage-recovery test.
FTA-QOS-003	Cellular-cost reporting shall account for retry and protocol overhead using the chosen network measurement boundary.	Modem/network counter correlation.

Verification, margin and future-change rule

Carrier plan, protocol, compression policy or uplink scheduler changes reopen bandwidth and cost-model evidence. Fleet cost optimization may not silently redefine protected data-retention requirements.

OFFLINE RETENTION, EMMC ENDURANCE & WEAR-LEVELING

Robust eMMC storage with wear-leveling supports offline buffering over the 15-year lifecycle intent. Endurance is derived from actual write traffic, buffering duty, filesystem amplification, temperature exposure and device-specific health behavior rather than from calendar time alone. The design records cumulative bytes written or equivalent product-level wear indicators where available and correlates them with fleet usage.

Retention policy can reduce avoidable writes by batching metadata and avoiding repeated rewrite of unchanged configuration. Storage reserve is maintained so wear-management and filesystem housekeeping are not forced to operate at a permanently full condition. Service diagnostics identify units approaching project-defined wear or capacity thresholds; exact P/E-cycle and data-retention limits remain controlled supplier/program parameters until released.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-WEAR-001	The eMMC endurance argument shall use measured or justified write amplification and the released fleet buffering mission profile.	Endurance model with bench write-traffic correlation.
FTA-WEAR-002	Storage health telemetry shall expose project-approved wear and capacity indicators without claiming unsupported remaining-life precision.	Field-diagnostic and device-health review.
FTA-WEAR-003	Retention and filesystem policies shall preserve reserve capacity required for controlled operation and maintenance.	Near-full storage and garbage-collection stress test.

Verification, margin and future-change rule

eMMC device, filesystem, logging frequency, compression ratio or fleet offline-duty assumption changes reopen the endurance model. A nominally larger device can have different endurance characteristics and requires evidence before substitution.

FILESYSTEM CORRUPTION, ABRUPT POWER LOSS & RECOVERY

Commercial vehicles can remove power without a software shutdown handshake. The persistent-data design therefore assumes abrupt interruption can occur at any write boundary. Recovery tests cut power during telemetry append, queue-index update, rule-package activation and software-update staging, then verify that the device either recovers valid prior state or identifies the affected data as incomplete.

Repair actions are bounded so a very large offline queue does not create an unacceptably long boot before acquisition can resume. Where full repair must continue in the background, the architecture protects new writes and avoids reusing uncertain storage regions until validation completes. Repeated corruption is escalated as a storage-health issue rather than repeatedly formatting the device and erasing fleet evidence without traceability.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-FS-001	Abrupt power loss during persistent writes shall not cause corrupted records or configuration to be accepted as valid.	Randomized power-cut campaign.
FTA-FS-002	Storage recovery shall have bounded startup impact and shall preserve diagnostic evidence of detected corruption.	Large-backlog recovery timing test.
FTA-FS-003	Automatic repair or reformat actions that can discard telemetry shall require released policy and auditable reason reporting.	Corruption-recovery policy test.

Verification, margin and future-change rule

Filesystem version, mount options, database/index implementation or storage size changes reopen power-cut and recovery-time evidence. A software update that changes on-disk format requires forward/backward compatibility or a controlled migration path.

DATA INTEGRITY, RECORD CHECKS & SCHEMA VALIDATION

Telemetry integrity is protected at several levels: CAN reception status, parser validity, local record structure, persistent-storage checks and authenticated cloud transport. These controls have different purposes and are not treated as interchangeable. A TLS-protected connection, for example, does not prove that a record was correctly decoded before transmission; similarly, a local checksum does not authenticate the cloud endpoint.

Records include length, type/version and integrity information appropriate to the storage format so truncated or misaligned blocks can be rejected independently. Backend ingestion validates schema and range rules before data is admitted to fleet analytics. When a field is unknown under the active decoder version, the system preserves the raw or opaque representation where configured rather than coercing it into a misleading value.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-INT-001	Persistent telemetry records shall include structural integrity information sufficient to detect truncation or corruption before decoding.	Corrupted-record injection test.
FTA-INT-002	Schema validation shall reject incompatible field versions without silently reinterpreting units or signedness.	Schema mismatch and boundary-value test.
FTA-INT-003	Transport security, record integrity and signal-validity status shall remain separately observable.	End-to-end fault classification test.

Verification, margin and future-change rule

Record format, checksum/hash, parser or cloud schema changes reopen corruption and compatibility tests. Integrity mechanisms shall be evaluated at the boundary they protect rather than credited generically across the whole telemetry chain.

DATA MINIMIZATION, VEHICLE IDENTIFIERS & PRIVACY BOUNDARIES

Fleet telemetry can contain identifiers, location-correlated information and operational history even when the gateway does not explicitly process personal data. The product therefore supports a configuration boundary for which identifiers and signals are collected, retained and transmitted. Data minimization is applied through fleet rules and schema selection rather than by inventing unsupported legal classifications in the PRD.

Unit serial number, vehicle association, modem identity and cloud tenant identity are separated where practical so service diagnostics do not expose more fleet information than required. Debug logs avoid storing private credentials or unrestricted payload dumps by default. Decommissioning and reassignment procedures remove or rebind fleet-specific credentials and cached configuration according to the owner-approved service process.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-PRV-001	The fleet configuration shall explicitly identify telemetry classes and identifiers permitted for collection and transmission.	Configuration and data-inventory review.
FTA-PRV-002	Diagnostic interfaces and logs shall not expose private keys or unrestricted credential material.	Diagnostic-access negative test.
FTA-PRV-003	Vehicle or fleet reassignment shall use a controlled process for clearing or rebinding retained configuration and credentials.	Decommission/reprovision service test.

Verification, margin and future-change rule

Schema, fleet identity model, log content or service process changes reopen data-inventory and decommissioning review. Regulatory applicability remains program-specific and is not inferred from generic fleet deployment.

REMOTE SOFTWARE / RULE UPDATE, ROLLBACK & VERSION COMPATIBILITY

Remote maintenance can update application software, fleet rules, signal dictionaries and cloud configuration, but each artifact has independent compatibility and rollback rules. The gateway stages updates in a way that preserves the last known qualified operational set until integrity and compatibility checks pass. A rule package requiring a newer parser cannot be activated merely because each package is individually signed.

Update scheduling accounts for vehicle power availability, storage reserve and connectivity. If the unit powers down during transfer or installation, the next boot either resumes safely or returns to the prior operational state. Update telemetry reports package identity, result and rollback reason without allowing management traffic to bypass the read-only vehicle-interface architecture.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-UPD-001	Software, signal-dictionary and fleet-rule packages shall be compatibility checked as a released configuration set before activation.	Mixed-version update matrix test.
FTA-UPD-002	Interrupted remote update shall preserve a bootable and configuration-consistent operational state.	Power/network interruption update test.
FTA-UPD-003	Update results and rollback reasons shall be auditable without exposing secret signing or credential material.	Update audit-log review.

Verification, margin and future-change rule

Package format, compatibility matrix, rollback partition or update transport changes reopen interrupted-update and mixed-version tests. A successful signature check does not establish semantic compatibility across independently versioned artifacts.

NETWORK SEGMENTATION, MAINTENANCE ACCESS & SECURITY INDEPENDENCE

Vehicle acquisition, cellular backhaul and maintenance interfaces occupy different trust roles. Network segmentation and process permissions prevent an exposed management service from gaining direct authority over protected credential storage or low-level vehicle-interface controls. Local service access is enabled only through released authentication and physical/service procedures, with debug features disabled or constrained in production configuration.

Security services are designed so their failure does not create a transmit-capable vehicle path. Conversely, read-only architecture does not excuse weak cloud security because compromised telemetry can damage fleet decisions or expose operational data. Maintenance commands are rate bounded and logged; a diagnostic session cannot consume enough CPU, storage or network bandwidth to starve normal acquisition without an explicit service mode.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-MNT-001	Maintenance and cloud-management paths shall be segmented from vehicle acquisition and protected credential functions according to the released security architecture.	Network/process permission review and penetration test.
FTA-MNT-002	Production debug access shall be disabled or controlled by the released service authorization process.	Production configuration audit.
FTA-MNT-003	Maintenance traffic shall have bounded resource consumption relative to high-priority vehicle acquisition.	Concurrent diagnostics and peak-traffic stress test.

Verification, margin and future-change rule

Service port, firewall, process permission, debug interface or maintenance workflow changes reopen segmentation and resource-isolation evidence. Field convenience features are not accepted if they create an undocumented path around production security controls.

9-36 V INPUT: CRANK, REVERSE POLARITY, SURGE & HOLD-UP DVP

The product-level electrical DVP decomposes the 9-36 V heavy-vehicle input envelope into program-controlled scenarios for low-voltage operation, crank, reverse connection, short interruption, surge and load-dump behavior. Each waveform, duration and source impedance shall come from the approved vehicle interface specification rather than being inferred from a generic example.

Testing monitors logic rails, reset, CAN bus disturbance, storage writes and modem behavior simultaneously. A power transient that does not damage hardware can still fail the product if it corrupts persistent queues or violates read-only bus behavior. Recovery criteria include deterministic boot, storage/configuration validation and a diagnostic record of the abnormal reset where energy remained sufficient to save it.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-TRN-001	Electrical transient DVP shall use the vehicle-program waveform set and source impedance for the released installation.	Approved transient matrix and lab records.
FTA-TRN-002	Transient exposure shall be evaluated for data integrity and vehicle-bus disturbance in addition to component survival.	Synchronized rail/bus/storage monitoring.
FTA-TRN-003	Post-transient restart shall validate active software, rule and storage state before service is declared healthy.	Recovery-state verification.

Verification, margin and future-change rule

Input protection, hold-up capacitance, regulator, storage or vehicle harness assumptions changes reopen only the affected waveform cases and recovery evidence. Qualification cannot be claimed from a nominal 9-36 V DC sweep alone.

THERMAL, HOT-SOAK, COLD-START & PASSIVE-COOLING DVP

Thermal qualification exercises the gateway at the released cab-mounted ambient and installation conditions with representative compute, storage and modem activity. Heat generation is workload dependent: cellular transmit bursts and sustained eMMC writes can coincide with CPU-heavy compression after a coverage outage. The test therefore includes steady acquisition, offline buffering, backlog upload and update/recovery modes rather than one artificial maximum-power loop.

Cold-start testing verifies power-up, oscillator behavior, storage mount and modem initialization without assuming all services become available simultaneously. Hot-soak testing records component and enclosure temperatures, throttling, queue growth and data completeness. If thermal protection sheds workload, the resulting telemetry gap or reduced upload rate is observable and the unit recovers without hidden configuration changes when temperature returns to range.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-THM-001	Thermal DVP shall cover the released workload combinations that can create peak compute, storage and modem dissipation.	Thermal chamber workload matrix.
FTA-THM-002	Cold-start qualification shall verify deterministic acquisition startup and storage/configuration integrity.	Cold-start repeated-cycle test.
FTA-THM-003	Thermal workload shedding shall be observable and shall not silently change fleet-rule configuration.	Overtemperature derating and recovery test.

Verification, margin and future-change rule

Enclosure, thermal pad, PCB copper, processor, modem or workload changes reopen the temperature cases they affect. Thermal equivalence requires measured path and workload evidence, not only matching maximum junction ratings.

HIGH-SHOCK CHASSIS, CONNECTOR / HARNESS & MOUNTING ROBUSTNESS

The supplied packaging baseline calls for a high-shock-resistant chassis mounted in commercial vehicle cabs. Mechanical qualification therefore links enclosure stiffness, PCB support, connector retention, eMMC and modem components, antenna leads and mounting fasteners to the actual installation spectrum. Cable mass and harness routing are included because they can dominate connector loads even when the enclosure itself survives the test.

Post-test acceptance covers more than power-up. Each CAN channel is checked for read-only electrical behavior, isolation integrity and intermittent receive errors; storage is checked for corruption and connector fretting symptoms; cellular performance is compared with pre-test baseline where the antenna path is affected. Fastener torque or mounting changes are controlled manufacturing characteristics where they influence structural response.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-MEC-001	Mechanical qualification shall use the released mounting points, fasteners, connector mates and representative harness masses.	Installation-representative vibration/shock test.
FTA-MEC-002	Post-stress acceptance shall include CAN acquisition, isolation, storage and backhaul functional checks.	Pre/post qualification comparison.
FTA-MEC-003	Critical mounting and connector assembly characteristics shall be controlled in manufacturing and service instructions.	Drawing/process audit and torque/retention verification.

Verification, margin and future-change rule

Enclosure alloy, bracket, connector, PCB support, harness mass or mounting orientation changes reopen structural correlation. Cosmetic similarity is not evidence of equivalent shock or fretting performance.

EMC / EMI, CAN-CELLULAR COEXISTENCE & SELF-INTERFERENCE

The aggregator combines sensitive CAN receivers, digital compute, switching power and a cellular transmitter in one product. EMC validation therefore evaluates both external immunity/emissions and internal coexistence. Modem transmit bursts are exercised while all CAN channels carry representative traffic and while eMMC writes occur, because coupling into transceiver reference, isolation supplies or clock domains can create intermittent data loss that is invisible in separate subsystem tests.

Conducted and radiated disturbances are monitored for false CAN frames, receive errors, storage corruption, processor resets and modem instability. The acceptance matrix distinguishes temporary telemetry loss with explicit diagnostics from persistent malfunction or vehicle-bus disturbance. Cable shields, chassis bond and antenna routing are treated as installation variables and are fixed for production-equivalent EMC evidence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-EMC-001	EMC DVP shall exercise cellular transmission concurrently with representative CAN acquisition and storage activity.	Combined immunity/emissions/coexistence test.
FTA-EMC-002	Electromagnetic disturbance shall not create intentional or sustained dominant drive on read-only vehicle interfaces.	Bus-state monitoring during immunity tests.
FTA-EMC-003	Production-equivalent EMC evidence shall identify cable, antenna, grounding and enclosure configuration.	Test-configuration audit.

Verification, margin and future-change rule

Modem, antenna, DC/DC, transceiver, cable routing, PCB stack or enclosure grounding changes reopen the affected emissions, immunity and coexistence evidence. Component compliance reports do not replace integrated-product testing.

MANUFACTURING / EOL - ISOLATED CAN, STORAGE & FUNCTIONAL TEST

End-of-line testing confirms the product was built as the released telemetry variant before fleet-specific configuration is applied. The fixture verifies supply current and startup, isolated CAN receive paths, absence of transmit authority, storage accessibility, modem identity, secure-boot state and critical diagnostics. Measurement limits and fixture uncertainty are controlled so EOL results can be compared across production lines and service replacements.

CAN testing uses known stimulus on every populated channel and checks channel identity through the recorded data path. A unit that can receive but has an unintended transmit-enabled path is rejected. Storage checks include basic read/write and health information without consuming excessive endurance during manufacturing. EOL failures are serialized and linked to rework history; a retested unit does not lose traceability to the original failure.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-EOL-001	EOL shall verify every populated vehicle channel for receive function and released read-only enforcement.	Automated CAN stimulus and negative TX-path test.
FTA-EOL-002	EOL shall confirm boot/security state, storage access, modem identity and critical hardware diagnostics before fleet provisioning.	Automated functional fixture record.
FTA-EOL-003	EOL measurement limits, equipment identity and rework history shall be traceable to the unit serial number.	Manufacturing genealogy audit.

Verification, margin and future-change rule

Fixture, test software, transceiver, secure-boot implementation or storage device changes reopen gauge correlation and affected EOL limits. Missing manufacturing evidence blocks unit release; it does not create a fictitious runtime safe state.

FLEET-RULE FLASHING, SERIALIZATION & CERTIFICATE PROVISIONING

After hardware EOL, the unit is provisioned with serialized identity, authorized device credentials and the fleet-specific rule/configuration package required for deployment. These steps are sequenced so credentials and rules are bound to the intended unit and tenant. Provisioning verifies artifact signatures, compatibility and resulting configuration identity rather than merely copying files to storage.

The manufacturing system records hardware revision, software image, rule package, signal dictionary, modem/SIM identity and credential issuance reference. Failed or interrupted provisioning leaves the unit in a non-deployable state until the process is repeated or formally recovered. Sensitive material is handled through controlled interfaces; production operators do not need direct access to private keys to complete provisioning.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-PRO-001	Fleet rules and credentials shall be provisioned only to a unit whose serialized hardware/software compatibility has been validated.	Provisioning workflow and mismatch test.
FTA-PRO-002	Provisioning genealogy shall record software, fleet-rule, signal-dictionary and credential identities associated with each serial number.	Manufacturing database audit.
FTA-PRO-003	Interrupted or failed provisioning shall prevent deployment until a controlled recovery process establishes a complete configuration.	Power/network interruption provisioning test.

Verification, margin and future-change rule

Provisioning station, certificate authority, rule package or manufacturing database changes reopen identity-binding and recovery evidence. Production equivalence requires both functional EOL completion and successful serialized provisioning.

SYSTEM DVP - READ-ONLY SAFETY, DATA PIPELINE & OUTAGE RECOVERY

System verification combines the distinctive product claims rather than testing each subsystem only in isolation. Representative heavy-vehicle CAN/J1939 traffic is applied across all released channels while filtering, compression, storage and cellular communication run concurrently. The campaign includes network silence, bursts, transport-protocol sessions, malformed frames, cellular outage, storage near-full conditions, power interruption and remote configuration changes.

The primary safety/assurance objective is zero intentional interference with vehicle dynamics from read-only interfaces. Data-quality objectives cover capture completeness, overflow accounting, timestamp semantics, schema fidelity and store-and-forward recovery. Cybersecurity tests demonstrate TLS mutual authentication and authorized management without crediting security as the mechanism that enforces read-only bus behavior. Release evidence identifies the exact hardware, software, rules and backend versions used.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-DVP-001	System DVP shall verify read-only vehicle-interface behavior under normal, overload, restart, update and fault conditions.	Integrated bus analyzer and fault campaign.
FTA-DVP-002	System DVP shall quantify capture loss and store-and-forward behavior under the released worst-case traffic, storage and connectivity combinations.	End-to-end telemetry completeness report.
FTA-DVP-003	Release evidence shall identify production-equivalent hardware, software, rule, credential and backend configurations.	DVP configuration audit and sign-off.

Verification, margin and future-change rule

Changes affecting interface hardware, scheduler, rule engine, storage, backhaul protocol or credential model reopen the integrated scenarios whose assumptions change. Incomplete critical DVP keeps the verification gate open and prevents a production-equivalence claim.

15-YEAR LIFECYCLE INTENT, FIELD DIAGNOSTICS & SECOND-SOURCE RELEASE

The product has a 15-year lifecycle intent and uses robust eMMC wear-leveling for offline buffering. Lifecycle evidence therefore tracks storage wear, reset history, CAN-channel faults, modem connectivity, software/rule version, credential status and recurring thermal or power events. These counters support maintenance and trend analysis but do not create unsupported predictions of exact remaining life.

Long-term support also covers secure update, credential renewal, fleet-rule evolution, eMMC and modem obsolescence, and second-source component introduction. A replacement part is not approved solely because pinout or capacity matches; the impact review considers bus loading, isolation, storage endurance/power-loss behavior, compute timing, RF performance, security and manufacturing test coverage. Field data can prioritize regression but cannot waive required qualification evidence.

Atomic product requirements

Requirement ID	Single-intent requirement	Verification / release evidence
FTA-LIF-001	Lifecycle diagnostics shall retain project-approved indicators for storage wear, reset history, interface health, software/rule identity and credential state.	Service-diagnostic and aging-counter review.
FTA-LIF-002	Second-source changes shall undergo similarity analysis and targeted regression for every affected electrical, timing, storage, RF, security and manufacturing assumption.	Supplier-change impact record and regression evidence.
FTA-LIF-003	A second source or lifecycle replacement shall not be approved for production equivalence until affected release evidence is closed.	Configuration-control and release-gate audit.

Verification, margin and future-change rule

Lifecycle, fleet-duty or supplier changes trigger a documented delta/impact review. Regression depth is proportional to the assumptions invalidated by the change; missing similarity or qualification evidence blocks the affected variant rather than being represented as a runtime ECU fault.